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QUANTITATIVE FINANCE

The Price of Being in a Hurry

A large order cannot be filled at the price that prompted it, and almost everything interesting about modern trading is an argument over how much of the difference was avoidable.

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In 1988 three researchers worked through fourteen thousand institutional trades on the New York Stock Exchange and added up what they had cost. The total came to twenty-three basis points of the money involved — a little under a quarter of one per cent. Eighteen of those basis points were commission. The other five were everything else: the drift in the price between the moment a manager decided to buy and the moment the shares actually arrived [1].

The five were the interesting part. To measure them at all, Stephen Berkowitz, Dennis Logue and Eugene Noser needed something to compare an execution against, and they chose the volume-weighted average price of the whole day's trading in that stock. If everyone else got 40.12 dollars and you got 40.19, you had paid seven cents for the privilege of being you. The benchmark stuck, and thirty-eight years later it is still the yardstick against which a large fraction of the world's institutional equity flow is judged.

The same year, André Perold gave the whole gap a name. A portfolio on paper, he pointed out, buys at the price on the screen at the moment of the decision; a real portfolio buys later, in pieces, at prices that have moved. The difference between the two is the *implementation shortfall*, and Perold's argument was that it is not an accounting nuisance to be minimised at the margin but the single largest reason paper portfolios outperform real ones [2].

Commissions have since collapsed to something close to zero. The five basis points did not follow them down. They could not, because they are not a fee for a service. They are the price of information.

The Price of Order Flow

Suppose you are a market maker and someone hands you a buy order for fifty thousand shares. You cannot see who they are. They might be a pension fund rebalancing on a rule written six months ago, in which case filling them is free money. They might also know something you do not, in which case filling them is a loss you have not noticed yet. You cannot distinguish the two cases, so you must charge a price that survives both.

Albert Kyle formalised this in 1985 in what remains the most-used model in market microstructure [3]. Three kinds of participant trade a single asset. An insider knows the asset's liquidation value v , drawn from a normal distribution around the prior price p_0 with variance Σ_0 . Noise traders submit a net order

u , normally distributed with standard deviation σ_u , for reasons unconnected to value. A competitive market maker sees only the total order flow $y = x + u$ and, earning zero expected profit, must set the price to the conditional expectation of v given y .

Look for an equilibrium in which both sides behave linearly: the insider trades $x = \beta(v - p_0)$ and the market maker quotes $p = p_0 + \lambda y$. The insider, facing that price rule, maximises $(v - p_0 - \lambda x)x$, which gives $x = (v - p_0)/2\lambda$ and hence $\beta = 1/2\lambda$. The market maker, running a regression of v on y , needs $\lambda = \beta\Sigma_0/(\beta^2\Sigma_0 + \sigma_u^2)$. Solving the pair:

$$\beta = \frac{\sigma_u}{\sqrt{\Sigma_0}}, \quad \lambda = \frac{\sqrt{\Sigma_0}}{2\sigma_u}, \quad \lambda\beta = \frac{1}{2}.$$

That λ — universally called Kyle's lambda — is the price impact of order flow, and it is the whole model in one number. It rises with the amount of private information floating about and falls with the volume of uninformed trading. Depth is not a property of the exchange; it is a property of how much camouflage the informed have to hide behind.

Three consequences follow immediately, and all three are exact. The insider's expected profit is $\frac{1}{2}\sigma_u\sqrt{\Sigma_0}$ and the noise traders lose precisely that amount — the market maker is a conduit, not a counterparty. The variance of the pricing error falls from Σ_0 to $\Sigma_0/2$: exactly half the private information ends up in the price, no more, because the insider throttles their own trading to keep the other half. And $\lambda\beta = 1/2$ says the same thing from the other side. The price moves halfway to the truth.

Put numbers on it. Take a fifty-dollar stock about which the informed party's edge is worth five dollars in standard deviation, against noise-trader flow with a standard deviation of two hundred thousand shares. Then β is forty thousand shares per dollar of mispricing and λ is 1.25 hundred-thousandths of a dollar per share, so a two-standard-deviation order imbalance of four hundred thousand shares moves the price by five dollars. The insider's expected haul is five hundred thousand dollars. A four-million-path Monte Carlo of the same equilibrium returns 499,407 dollars for the insider, 499,589 dollars of losses for the noise traders, and a posterior variance of 12.49 against the predicted 12.5.

IF PRICE IMPACT IS REALLY THE MARKET INFERRING YOUR INFORMATION, WHY SHOULD AN INDEX FUND WITH NO INFORMATION PAY IT?

Because the market maker cannot tell. That is the entire content of Lawrence Glosten and Paul Milgrom's companion paper of the same year, which derives the bid-ask spread from the same asymmetry rather than from inventory or processing costs [4]. Anonymity is symmetric: it protects the informed trader from being read, and it forces the uninformed trader to pay as though they might be. An index fund's impact cost is a tax levied on it for resembling somebody else.

The Part You Cannot Avoid

Kyle explains why impact exists. It does not tell a trader with a million shares to sell what to do on Tuesday morning. That problem was posed properly by Dimitris Bertsimas and Andrew Lo in 1998, who solved it by dynamic programming for a risk-neutral trader and found the answer to be dull: with linear impact and no risk aversion, split the order into equal pieces [5]. Equal pieces are also what a naive trader does. The model's own logic had made the model uninteresting.

Robert Almgren and Neil Chriss fixed this in 2000 by noticing what was missing [6]. A trader who spreads an order over five days is not indifferent between the possible outcomes; they are exposed to five days of price movement, which is a risk they did not want and were not paid for. Minimising expected cost alone will always recommend trading slowly, all the way to infinitely slowly. Something has to push back.

Their framework separates impact into two channels. Selling n_k shares in period k of length τ depresses the price permanently by γn_k — that part never comes back, and every subsequent share is sold into the lower price. It also depresses the execution price of that slice alone by $\eta n_k / \tau$, proportional to the *rate* of trading, which recovers as soon as you stop. Add half a bid-ask spread ϵ per share, let the holdings path be $x_0 = X$ down to $x_N = 0$, and the two moments of the shortfall are

$$E = \frac{1}{2} \gamma X^2 + \epsilon \sum_k |n_k| + \frac{\tilde{\eta}}{\tau} \sum_k n_k^2, \quad V = \sigma^2 \tau \sum_k x_k^2,$$

with $\tilde{\eta} = \eta - \gamma\tau/2$. Now look hard at the first term of E . It contains no n_k at all. Whatever schedule you choose, the permanent damage costs $\gamma X^2/2$ — because the price ends up γX lower no matter what, and on average you sold into half of that decline. Sell the whole position in one second or dribble it out over a month: the permanent component is identical to the penny.

This is the model's most useful result and it is almost never stated as one. Linear permanent impact is a sunk cost of having the position at all. Every decision an execution algorithm makes — every argument about urgency, participation rate and schedule — is an argument about the *other* two terms. The market's memory of your trade is not something you can trade your way out of.

The Shape of Haste

What remains is a genuine trade-off, and it has a closed form. Minimise $E + \lambda V$, where λ measures how many dollars of expected cost the trader will pay to remove one dollar-squared of variance, and the optimal holdings follow

$$x(t) = X \frac{\sinh(\kappa(T-t))}{\sinh(\kappa T)}, \quad \cosh(\kappa\tau) = 1 + \frac{\tau^2}{2} \frac{\lambda\sigma^2}{\tilde{\eta}}.$$

A hyperbolic sine is an exponential decay bent to hit zero exactly at the deadline, and the number that matters is the trading half-life $\ln 2/\kappa$. It compresses volatility, liquidity and nerve into a single figure with units of days, and a trader who knows it knows the whole schedule.

Almgren and Chriss's own worked example is a million shares of a fifty-dollar stock whose average daily volume is five million shares, to be liquidated over five days. Their impact coefficients look arbitrary until you check what they encode: $\gamma = 2.5 \times 10^{-7}$ means that trading ten per cent of a day's volume moves the price permanently by 0.125 dollars, which is exactly the assumed bid-ask spread of an eighth; $\eta = 2.5 \times 10^{-6}$ means that trading one per cent of daily volume in a day costs the same eighth temporarily. The volatility of 0.95 dollars a day is thirty per cent a year on a fifty-dollar stock. Every one of those calibrations reproduces to the last digit.

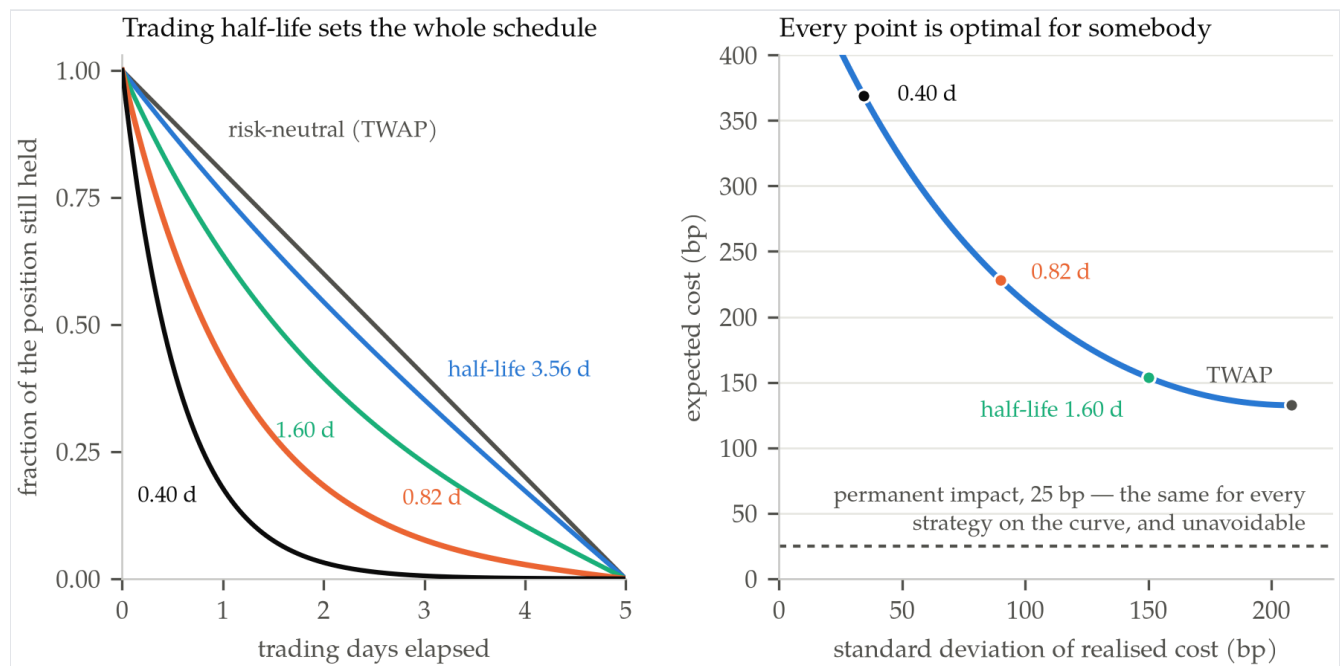


FIGURE Left: optimal liquidation paths for the Almgren-Chriss worked example, labelled by trading half-life; the risk-neutral limit is the straight line. Right: the same strategies as points on the efficient frontier of execution, in basis points of the 50-million-dollar position.

Risk aversion (per million dollars)	Trading half-life	Expected cost (bp)	Cost s.d. (bp)
0 (risk-neutral)	infinite	132	208
0.1	3.56 days	134	192
0.5	1.60 days	154	150
2	0.82 days	228	90
10	0.40 days	369	34
unbounded	0	513	0

Read the table as a menu rather than a ranking. The risk-neutral trader accepts an expected cost of 132 basis points with a standard deviation of 208 — the outcome is more uncertain than it is expensive, which is the fact that makes the problem worth solving. Buying the first tranche of certainty is nearly free: the second row cuts the standard deviation by sixteen basis points for a cost of two. Buying the last tranche is ruinous. Twenty-five of those basis points, in every single row, are the permanent-impact term, unchanged and unchangeable.

The risk-neutral limit deserves its own line. As $\lambda \rightarrow 0$, $\kappa \rightarrow 0$ and the hyperbolic sine straightens into $X(1 - t/T)$: equal slices, which is exactly Bertsimas and Lo's answer and exactly what a time-weighted average price algorithm does. TWAP is not a lazy default. It is the optimal strategy for a trader who does not care about risk, which is a coherent thing to be and a rare thing to admit.

One caveat that the elegance conceals. The optimal path is *deterministic*: it is fixed before the first share trades and never consults the price. Almgren returned to this with Julian Lorenz in 2007 and showed that a trader who accelerates when the price moves in their favour can do materially better, even in a market with no predictability whatsoever, because the mean-variance objective is not itself dynamically consistent [7]. The static schedule is a very good first answer, not a final one.

Why the Impact Has to Be Linear

Assuming permanent impact is linear in size looks like the sort of convenience a modeller adopts to get a closed form. It is not. It is forced.

Gur Huberman and Werner Stanzl proved in 2004 that if the permanent impact of a trade is time-independent, then any shape other than a straight line admits *quasi-arbitrage* — a round-trip sequence of trades with positive expected profit and no position at the end [8]. Convex impact lets you profit by buying in small pieces and selling in one large one; concave impact lets you profit by doing the reverse. Only linearity makes every round trip cost money. Their result also grants the licence Almgren and Chriss take: once trades have a temporary impact as well, only the permanent part must be linear, and the temporary part is free to be any reasonable shape.

Jim Gatheral pushed the same argument into the time domain in 2010, asking not what impact does but how it decays [9]. If the impact of a trade decays like $G(\tau) \sim \tau^{-\gamma}$ and instantaneous impact scales like v^δ then the requirement that round trips cost money constrains the pair of exponents. Exponential decay, the assumption most convenient for stochastic control, turns out to be compatible only with linear impact — which is to say that the two most popular simplifying assumptions in the field are incompatible with each other unless you make both. Empirically, γ comes out near 0.4 and δ near 0.6, and their sum is close to one: markets sit approximately on the no-arbitrage boundary rather than comfortably inside it.

That decay is not a modelling artefact either. Anna Obizhaeva and Jiang Wang rebuilt the problem in 2013 around a limit order book that is eaten by a trade and then refills at a finite rate, and found that the optimal strategy changes shape entirely [10]. It begins with a discrete block — deliberately pushing the

book away from its resting state to attract replenishment — then trades continuously, then finishes with another block. What governs the schedule is *resilience*, the speed at which the book heals, not the static depth or spread that a snapshot would show you.

The Square Root and the Size of a Fund

Theory says linear. Data says otherwise, and has for thirty years.

The empirical regularity, first documented in BARRA's internal impact handbook in 1997 [11] and confirmed on every instrument anyone has looked at since, is that executing a large order of Q shares in a market that trades V shares a day moves the price by roughly

$$I(Q) \approx Y \sigma \sqrt{Q/V},$$

with σ the daily volatility and Y a constant of order one. Almgren and colleagues estimated it directly from nearly seven hundred thousand Citigroup orders spanning December 2001 to June 2003, and rejected the square root in favour of a $3/5$ power for the temporary component, while being unable to reject linearity for the permanent one [12]. Bence Tóth and co-authors then showed the same law holding across futures, equities and eventually cryptocurrencies, over decades and continents [13]. A relationship that survives that much variation is telling you about something structural — in their reading, that the visible liquidity in a market is a vanishing fraction of the latent liquidity, and a large order is always trading against a book that mostly does not exist yet.

The square root has one consequence that ought to be better known. If impact grows as $\sqrt{Q}$, then the total dollar cost of an order grows as $Q^{3/2}$: doubling your size multiplies your execution bill by 2.83. Now suppose a strategy's gross edge is proportional to the money deployed, aQ , while its cost is $bQ^{3/2}$. Net profit $aQ - bQ^{3/2}$ is maximised at

$$Q^* = \frac{4a^2}{9b^2}, \quad \frac{\text{cost at } Q^*}{\text{gross alpha at } Q^*} = \frac{2}{3}.$$

At its own optimal size, a strategy pays away exactly two-thirds of its gross edge in market impact and keeps one third. The ratio does not depend on how good the signal is, or on how liquid the market is: a better signal and a deeper market both buy you a larger Q^* , not a better split. This is the arithmetic behind the otherwise puzzling behaviour of successful quantitative funds — that they close to new money at a size which looks arbitrary from outside, and that a manager's honest answer to "how much can you run?" is a statement about b , not about skill.

How large is the bill in practice? Estimates diverge sharply depending on whose trades are being measured. Andrea Frazzini, Ronen Israel and Tobias Moskowitz examined 1.7 trillion dollars of live executions across twenty-one developed equity markets over nineteen years and found real costs roughly an order of magnitude below what the academic literature, working from daily or intraday price data, had inferred — a median of about six basis points per rebalance [14]. The gap is not a contradiction. A

patient trader with good execution technology pays much less than a naive one, which is the entire justification for the industry that Almgren and Chriss's paper helped create, and Robert Engle, Robert Ferstenberg and Jeffrey Russell's econometric treatment of cost and risk jointly is the natural way to compare the two [15].

Nine Per Cent of the Previous Minute

At 2:32 in the afternoon on 6 May 2010, a mutual fund complex began selling 75,000 E-Mini S&P 500 futures contracts, worth about 4.1 billion dollars, as a hedge against an existing equity position. It used an automated algorithm configured to execute at nine per cent of the trading volume of the previous minute, without regard to price or time [16].

Nine per cent of trailing volume is a defensible rule. It is the practical form of the intuition that impact depends on participation rate, and on a normal day it is close to what an Almgren-Chriss schedule would do. But it contains a feedback loop that the framework does not model: it treats market volume as exogenous, when volume is partly a response to your own selling. As high-frequency market makers absorbed the flow and passed contracts rapidly among themselves, volume rose; the algorithm read rising volume as rising liquidity and accelerated into it. The 75,000 contracts went out in about twenty minutes, a pace which at nine per cent participation implies some 830,000 contracts changing hands alongside them; an order of that size would ordinarily be spread over several hours. Across the whole day the fund accounted for a little over one per cent of E-Mini trading. Across the minutes that mattered it was nine.

The two headline figures are at least internally consistent: 4.1 billion dollars across 75,000 contracts, at the E-Mini's fifty-dollar multiplier, implies an average execution near 1,093 index points. The staffs of the Commodity Futures Trading Commission and the Securities and Exchange Commission published their reconstruction that September, and Andrei Kirilenko, Albert Kyle, Mehrdad Samadi and Tugkan Tuzun later worked through the audit-trail data account by account [17]. Their conclusion was not that high-frequency traders caused the crash, but something more uncomfortable: the intermediaries behaved exactly as inventory-constrained market makers should, and the constraint bound at the worst possible moment.

There is a reading of that afternoon which is really a reading of everything above. Kyle's model says that the market cannot tell an uninformed seller from an informed one, and prices order flow accordingly. A participation-rate algorithm is a machine for converting a fund's private urgency into public order flow at a fixed rate. On 6 May the market read four billion dollars of hedging as information, and priced it as though it were. The mathematics of optimal execution is the discipline of arranging your own trades so that the inference drawn from them is as small, and as expensive to act on, as you can afford to make it. It is not a way of avoiding the cost of being in a hurry. It is a way of finding out what that cost is before you pay it.

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